17.4: Translation is the RNA-directed synthesis of a polypeptide: *A Closer Look*

We will now examine how genetic information flows from mRNA to protein—the process of translation **(Figure 17.15)**. We'll focus on the basic steps of translation that occur in both bacteria and eukaryotes, while pointing out key differences.

Figure 17.15

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Translation: the basic concept.
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m R N A inside ribosome. Transfer R N A binds to m R N A codon and transfers a Phe to growing polypeptide chain. Other t R N As preparing to bind to m R N A codons and transfer their amino acids.

As a molecule of mRNA is moved through a ribosome, codons are translated into amino acids, one by one. The translators, or interpreters, are tRNA molecules, each type with a specific anticodon at one end and a corresponding amino acid at the other end. A tRNA adds its amino acid cargo to a growing polypeptide chain when the anticodon hydrogen-bonds to the complementary codon on the mRNA.

Overview: In the nucleus, transcription of D N A results in pre-m R N A. R N A processing leads to m R N A, which leaves the nucleus. Translation of the m R N A takes place in the cytoplasm on ribosomes, where polypeptides are synthesized. Messenger R N A is inside ribosome. Various amino acids are drawn present in the cytoplasm and some are already attached to different t R N A molecules. Here, t R N As are drawn as upside-down L-shaped molecules with triplet anticodons on one end and attachment points for amino acids on the other end. Messenger R N A nucleotide sequence shown is 5 prime-U G G U U U G G C-3 prime. Arrows indicate that m R N A will slide to the left as translation proceeds. A t R N A has bound to the m R N A codon U U U inside the ribosome with the t R N A anticodon A A A. This t R N A is transferring a Phe amino acid to the growing polypeptide chain. The t R N A that has already transferred a T r p to the U G G m R N A codon (left of U U U) is being released from the ribosome. This t R N A has an anticodon reading A C C. Another t R N A carrying a G l y is about to bind to the m R N A codon G G C (right of U U U) with anticodon G G C.

BioFlix® Animation

Translation

image

image

Animation

Overview of Translation

image

image

Molecular Components of Translation

In the process of translation, a cell “reads” a genetic message and builds a polypeptide accordingly. The message is a series of codons along an mRNA molecule, and the translator is called a **transfer RNA (tRNA)**. The function of a tRNA is to transfer an amino acid from the cytoplasmic pool of amino acids to a growing polypeptide in a ribosome. A cell keeps its cytoplasm stocked with all 20 amino acids, either by synthesizing them from other compounds or by taking them up from the surrounding solution. The ribosome, a structure made of proteins and RNAs, adds each amino acid brought to it by a tRNA to the growing end of a polypeptide chain (see Figure 17.15).

Translation is simple in principle but complex in its biochemistry and mechanics, especially in the eukaryotic cell. In dissecting translation, we’ll focus on the slightly less complicated version of the process that occurs in bacteria. We’ll first look at the major players in this process.

The Structure and Function of Transfer RNA

The key to translating a genetic message into a specific amino acid sequence is the fact that each tRNA molecule enables translation of a given mRNA codon into a certain amino acid. This is possible because a tRNA bears a specific amino acid at one end of its three-dimensional structure, while at the other end is a nucleotide triplet that can base-pair with the complementary codon on mRNA.

A tRNA molecule consists of a single RNA strand that is only about 80 nucleotides long (compared to hundreds of nucleotides for most mRNA molecules). The presence of complementary stretches of nucleotide bases that can hydrogen-bond to each other allows this single strand to fold back on itself and form a molecule with a particular 3-D structure. Flattened into one plane to clarify this base pairing, a tRNA molecule looks like a cloverleaf **(Figure 17.16a)**. The tRNA actually twists and folds into a compact 3-D structure that is roughly L-shaped **(Figure 17.16b)**, with the 5′ and 3′ ends of the linear tRNA both located near one end of the structure. The protruding 3′ end acts as the attachment site for an amino acid. The loop

extending from the other end of the L includes the **anticodon**, the particular nucleotide triplet that base-pairs to a specific mRNA codon. Thus, the structure of a tRNA molecule fits its function.

Figure 17.16

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The structure of transfer RNA (tRNA).
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The structure of transfer R N A (t R N A).

The parts are as follows.

•Part a. Two-dimensional structure. The four base-paired regions and three loops are characteristic of all t R N As, as is the base sequence of the amino acid attachment site at the 3prime end. The anticodon triplet is unique to each t R N A type, as are some sequences in the other two loops. (The asterisks mark bases that have been chemically modified, a characteristic of t R N A. The modified bases contribute to t R N A function in a way that is not yet understood.) Two-dimensional structure of t R N A consisting a clover-leaf shape with four base-paired regions and three loops. Loops are at the ends of three of the arms; there is no loop at the final end, where the 5 prime and 3 prime ends of the t R N A molecule are found. No H bonds form between bases in the loops. The amino acid attachment site is at the 3 prime end and the anticodon triplet (unique to each t R N A) is at the tip of the loop opposite the 5 and 3 prime ends. Anticodon shown is A A G. H-bonds (dashed lines) form between complementary base pairs (C and G, A and U) in the arms of the cloverleaf. One asterisk is drawn in the anticodon loop, three in a second loop and 2 in the third loop; these asterisks mark bases that have been chemically modified, a characteristic of t R N A. The modified bases contribute to t R N A function in a way that is not yet understood.

•Part b. Three-dimensional structure of t R N A as a ribbon model. Twisted R N A strand forms a rough upside-down L shape. Anticodon is three nucleotide bases in one loop; amino acid attachment site is the 3 prime end. Hydrogen bonds form between complementary base pairs.

•Part c. Transfer R N A symbol used in this text. Upside down L with amino acid attachment point at one end and anticodon (three bases) at the other. 3 prime end to left of anticodon and 5 prime end to right. Anticodon shown is A A G.

VISUAL SKILL *Look at the tRNA shown in this figure. Based on its anticodon, identify the codon it would bind to, as well as the amino acid that it would carry.*

*For suggested answer, see*Appendix A.**

HHMI BioInteractive Video

RNA Folding

image

image

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Anticodons are conventionally written 3′→5′ to align properly with codons written 5′→3′ (see Figure 17.15). (For base pairing, RNA strands must be antiparallel, like DNA.) As an example of how tRNAs work, consider the mRNA codon 5′-GGC-3′, which is translated as the amino acid glycine. The tRNA that base-pairs with this codon by hydrogen bonding has 3′-CCG-5′ as its anticodon and carries glycine at its other end (see the incoming tRNA approaching the ribosome in Figure 17.15). As an mRNA molecule is moved through a ribosome, glycine will be added to the polypeptide chain whenever the codon 5′-GGC-3′ is presented for translation. Codon by codon, the genetic message is translated as tRNAs position each amino acid in the order prescribed, and the ribosome adds that amino acid onto the growing polypeptide chain. The tRNA molecule is a translator in the sense that, in the context of the ribosome, it can read a nucleic acid word (the mRNA codon) and interpret it as a protein word (the amino acid).

Like mRNA and other types of cellular RNA, transfer RNA molecules are transcribed from DNA templates. In a eukaryotic cell, tRNA, like mRNA, is made in the nucleus and then travels to the cytoplasm, where it will participate in the process of translation. In both bacterial and eukaryotic cells, each tRNA molecule is used repeatedly, picking up its designated amino acid in the cytosol, depositing this cargo onto a polypeptide chain at the ribosome, and then leaving the ribosome, ready to pick up another of the same amino acid.

The accurate translation of a genetic message requires two instances of molecular recognition. First, a tRNA that binds to an mRNA codon specifying a particular amino acid must carry that amino acid, and no other, to the ribosome. The correct matching up of tRNA and amino acid is carried out by a family of related enzymes that are aptly named **aminoacyl-tRNA synthetases(Figure 17.17)**. The active site of each type of aminoacyl-tRNA synthetase fits only a specific combination of amino acid and tRNA. (Regions of both the amino acid attachment end and the anticodon end of the tRNA ensure the specific fit.) There are 20 different synthetases, one for each amino acid. A synthetase joins a given amino acid to an appropriate tRNA; one synthetase is able to bind to all the different tRNAs for its particular amino acid. The synthetase catalyzes the covalent attachment of the amino acid to its tRNA in a process driven by the hydrolysis of ATP. The resulting aminoacyl tRNA, also called a *charged tRNA*, is released from the enzyme and is then available to deliver its amino acid to a growing polypeptide chain on a ribosome.

Figure 17.17

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Aminoacyl-tRNA synthetases provide specificity in joining
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amino acids to their tRNAs.
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Aminoacyl-t R N A synthase links t R N A and its specific amino acid with covalent bonding using A T P.

Linkage of a tRNA to its amino acid is an endergonic process that occurs at the expense of ATP, which loses two phosphate groups, becoming AMP.

The steps are as follows.

1. The amino acid and the appropriate t R N A enter the active site of the specific synthase. Here, Tyrosine (Tyr) is the amino acid and the enzyme is Tyrosyl-t R N A synthase. This enzyme can only bind tyrosine and Tyr-t R N A, which carries the anticodon A U A. This anticodon is complementary to the Tyr codon on m R N A.
2. Using A T P (which goes to AMP plus to inorganic phosphate), the synthase catalyzes the covalent bonding of the amino acid to its specific t R N A. Here, Tyrosine is linked to Tyr-t R N A.
3. The t R N A, charged with its amino acid, is released by the synthase.

A computer model shows Aminoacyl-t R N A synthase as a globular enzyme bound to t R N A (twisted ribbon with nucleotides) and the amino acid it is attaching.

Animation

Transfer RNA

image

image

The second instance of molecular recognition is the pairing of the tRNA anticodon with the appropriate mRNA codon. If one tRNA type existed for each mRNA codon specifying an amino acid, there would be 61 tRNAs (see Figure 17.6). In bacteria, however, there are only about 45 tRNAs, signifying that some tRNAs must be able to bind to more than one codon. This is possible because the rules for base pairing between the third nucleotide base of a codon and the corresponding base of a tRNA anticodon are relaxed compared to those at other codon positions. For example, the nucleotide base U at the 5′ end of a tRNA anticodon can pair with either A or G in the third position (at the 3′ end) of an mRNA codon. The flexible base pairing at this codon position is called **wobble**. Wobble explains why the synonymous codons for a given amino acid most often differ in their third nucleotide base. Accordingly, a tRNA with the anticodon 3′-UCU-5′ can base-pair with either the mRNA codon 5′-AGA-3′ or 5′-AGG-3′, both of which code for arginine (see Figure 17.6).

The Structure and Function of Ribosomes

Although the ribosomes of bacteria and eukaryotes are very similar in structure and function, eukaryotic ribosomes are slightly larger and differ somewhat from bacterial ribosomes in their molecular composition. The differences are medically significant. Certain antibiotic drugs can inactivate bacterial ribosomes without affecting the ability of eukaryotic ribosomes to make proteins. These drugs, including tetracycline and streptomycin, are used to combat bacterial infections.

Ribosomes facilitate the specific coupling of tRNA anticodons with mRNA codons during protein synthesis. A ribosome consists of a large subunit and a small subunit, each made up of proteins and one or more **ribosomal RNAs**, or **rRNAs (Figure 17.18)**. In eukaryotes, the subunits are made in the nucleolus. Ribosomal RNA genes are transcribed, and the RNA is processed and assembled with proteins imported from the cytoplasm. Completed ribosomal subunits are then exported via nuclear pores to the cytoplasm. In both bacteria and eukaryotes, a large and a small subunit join to form a functional ribosome only when attached to an mRNA molecule. About one-third of the mass of a ribosome is made up of proteins; the rest consists of three rRNA molecules (in bacteria) or four (in eukaryotes). Because most cells contain thousands of ribosomes, rRNA is the most abundant type of cellular RNA.

Figure 17.18

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The anatomy of a functioning ribosome.
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The anatomy of a functioning ribosome.

The parts are as follows.

•Part a. Computer model of functioning ribosome. This is a model of a bacterial ribosome, showing its overall shape. The eukaryotic ribosome is roughly similar. Overview: In the nucleus, transcription of D N A results in pre-m R N A. R N A processing leads to m R N A, which leaves the nucleus. Translation of the m R N A takes place in the cytoplasm on ribosomes, where polypeptides are synthesized. A ribosomal subunit is a complex of ribosomal R N A molecules and proteins. This model shows two globular subunits; the large subunit sits above the small subunit. A strand of m R N A sits in the ribosome; t R N A molecules transfer amino acids within the ribosome. Three sites, E, P and A, are labeled within the ribosome. The growing polypeptide chain is seen protruding from an exit tunnel in the ribosome.

•Part b. Schematic model showing binding sites. Ribosome has m R N A binding site and three t R N A binding sites called A, P and E sites. This schematic ribosome will appear in later diagrams. Ribosome with large upper subunit and small lower subunit. Components of the ribosome labeled as follows: Small subunit:

m R N A binding site, near the top of the small subunit. Large subunit: Three t R N A binding sites drawn in middle region of upper subunit; from left to right: E (Exit) site; P (Peptidyl-t R N A) binding site; A (Aminoacyl-t R N A binding site). The exit tunnel is drawn above the P site.

•Part c. Schematic model with m R N A and t R N A. A t R N A fits into a binding site when its anticodon base-pairs with an m R N A codon. An m R N A is shown bound between the ribosomal subunits with the 5 prime end on the left and the 3 prime end on the right. The P site holds the t R N A attached to the growing polypeptide. The A site holds the t R N A carrying the next amino acid to be added to the polypeptide chain. Discharged t R N As leave from the E site. The polypeptide grows at its carboxyl end (opposite the amino end).

Animation

Ribosomes

image

image

The structure of a ribosome reflects its function of bringing an mRNA molecule together with tRNAs carrying amino acids. The mRNA itself has a binding site for the ribosome. (In the **Scientific Skills Exercise**, you can work with DNA sequences representing this binding site in a group of *Escherichia coli* genes.) The ribosome, in turn, has a binding site for mRNA, as well as three binding sites for tRNA (see Figure 17.18). The **P site** (peptidyl-tRNA binding site) holds the tRNA carrying the growing polypeptide chain, while the **A site** (aminoacyl-tRNA binding site) holds the tRNA carrying the next amino acid to be added to the chain. Discharged tRNAs leave the ribosome from the **E site** (exit site). The ribosome holds the tRNA and mRNA in close proximity and positions the new amino acid so that it can be added to the carboxyl end of the growing polypeptide. It then catalyzes the formation of the peptide bond. As the polypeptide becomes longer, it passes through an *exit tunnel* in the ribosome's large subunit. When the polypeptide is complete, it is released through the exit tunnel.

Scientific Skills Exercise

Interpreting a Sequence Logo

AP Science Practices 2, 4, 5, 6

How Can a Sequence Logo Be Used to Identify Ribosome Binding Sites on Bacterial mRNAs? When initiating translation, ribosomes bind to an mRNA at a ribosome binding site upstream of the AUG start codon. Because mRNAs from different genes all bind to a ribosome, the genes encoding these mRNAs are likely to have a similar base sequence where the ribosomes bind. Therefore, candidate ribosome binding sites on mRNA can be identified by comparing DNA sequences (and thus the mRNA sequences) of multiple genes in a species, searching the region upstream of the start codon for shared (conserved) stretches of bases. In this exercise, you will analyze DNA sequences from multiple such genes, represented by a visual graphic called a sequence logo.

The ribosome binding site on m R N A is toward the 5 prime end.

How the Experiment Was Done The DNA sequences of 149 genes from the *E. coli* genome were aligned using computer software. The aim was to identify similar base sequences—at the appropriate location in each gene—as potential ribosome binding sites. Rather than presenting the data as a series of 149 sequences aligned in a column (a sequence alignment), the researchers used a sequence logo.

Data from the Experiment To show how sequence logos are made, the potential ribosome binding regions from ten *E. coli* genes are shown below in a sequence alignment, followed by the sequence logo derived from the aligned sequences. Note that the DNA shown is the nontemplate (coding) strand, which is how DNA sequences are typically presented. (All data from Thomas D. Schneider.)

Sequence logo derived from the aligned sequences of 10 E. coli genes.

Sequence alignment

Sequence logo derived from the aligned sequences of 10 E. coli genes. Horizontal-axis is 5 prime to 3 prime left to right and is primary sequence of D N A by nucleotide position with values from negative 18 (left) to 8 (right) with increments every whole number. Nominal size is from 1 to 15. Data from the logo includes the following from top to bottom: negative 15 (A G C, total height less than 1); negative 14 (A T C G, total height less than 1); negative 13 (C (1.5) A (1) G, total height less than 3); negative 10 (G (1.5) A (1.5) C T, total less than 4); negative 9 (G (4), A (1), C (1), total height 6); negative 8 (G (1) A T, total height 2); negative 7 (A (1.5), G (1.5), T (1)); negative 3 (A G, total height less than 1); negative 1 (T (0.7) A (0.5) C (0.7)); 0 (A (5.5), G(1), T(1)); 1 (T (15)); 2 (G(15)); 3 (A C, total height less than 0.5); 4 (C (1.7) A (1.7) G (.4)); 7 (T (.7) A (.7) C (.6)).

Letters are placed on a horizontal axis labeled 5 prime to 3 prime from left to right. The tallest part of the sequence is A G T at 0, T at 1, and G at 2.

Sequence logo

Further Reading T. D. Schneider and R. M. Stephens, Sequence logos: A new way to display consensus sequences, *Nucleic Acids Research*18:6097–6100 (1990).

INTERPRET THE DATA

- In the sequence logo, the horizontal axis shows the primary sequence of the DNA by nucleotide position. Letters for each base are stacked on top of each other according to their relative frequency at that position among the aligned sequences, with the most common base as the largest letter at the top of the stack. The height of each letter represents the relative frequency of that base *at that position*.
- In the sequence alignment, count the number of each base at position −9 and order them from most to least frequent. Compare this to the size and placement of each base at −9 in the logo.
- Do the same for positions 0 and 1.
- The height of a stack of letters in a logo indicates the predictive power of that stack (determined statistically). If the stack is tall, we can be more confident in predicting what base will be in that position if a new sequence is added to the logo. For example, at position 2 in the sequence alignment, all ten sequences have a G; the probability of finding a G there in a new sequence is very high, as is the stack in the sequence logo. For short stacks, the bases all have about the same frequency, so it's hard to predict a base at those positions.
- Looking at the sequence logo, which two positions have the most predictable bases? What bases do you predict would be at those positions in a newly sequenced gene?
- Which 12 positions have the least predictable bases? How do you know? How does this reflect the relative frequencies of the bases shown at these positions in the sequence alignment? Use the two leftmost positions of the 12 as examples in your answer.
- In the actual experiment, the researchers used 149 sequences to build their sequence logo, which is shown below. There is a stack at each position, even if short, because the sequence logo includes more data.
- Which three positions in this sequence logo have the most predictable bases? Name the most frequent base at each.
- Which four positions have the least predictable bases? How can you tell?

Letters are placed on a horizontal axis labeled 5 prime to 3 prime from left to right. The tallest part of the sequence is A G T at 0, T at 1, and G at 2.

- A consensus sequence identifies the base occurring most often at each position in the set of sequences.
- Write out the consensus sequence of this (the nontemplate) strand. In any position where the base can't be determined, put a dash.
- Which provides more information—the consensus sequence or the sequence logo? What is lost in the less informative method?
-
    - Based on the logo, what five adjacent base positions in the 5′ UTR region are most likely to be involved in ribosome binding? Explain.
- What is represented by the bases in positions 0–2?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

The widely accepted model is that rRNAs, rather than ribosomal proteins, are primarily responsible for both the structure and the function of the ribosome. The proteins, which are largely on the exterior, support the shape changes of the rRNA molecules as they carry out catalysis during translation. Ribosomal RNA is the main constituent of the A and P sites and of the interface between the two subunits; it also acts as the catalyst of peptide bond formation. Thus, a ribosome could actually be considered one colossal ribozyme!

Building a Polypeptide

We can divide translation, the synthesis of a polypeptide, into three stages: initiation, elongation, and termination. All three require protein "factors" that aid in the translation process. Some steps of initiation and elongation also require energy, provided by the hydrolysis of guanosine triphosphate (GTP).

Ribosome Association and Initiation of Translation

The initiation stage of translation brings together an mRNA, a tRNA bearing the first amino acid of the polypeptide, and the two subunits of a ribosome. First, a small ribosomal subunit binds to both the mRNA and a specific initiator tRNA, which carries the amino acid methionine. In bacteria, the small subunit can bind the two in either order; it binds the mRNA at a specific RNA sequence, just upstream of the AUG start codon. In eukaryotes, the small subunit, with the initiator tRNA already bound, binds to the 5′ cap of the mRNA and then moves, or *scans*, downstream along the mRNA until it reaches the AUG start codon, where the initiator tRNA then hydrogen-bonds, as shown in step 1 of **Figure 17.19**. In either bacteria or eukaryotes, the AUG signals the start of translation; this is important because it establishes the codon reading frame for the mRNA.

Figure 17.19

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The initiation of translation.
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The initiation of translation.

1. Small ribosomal subunit binds to m R N A (5 prime end to the left and 3 prime end to the right). In a bacterial cell, the m R N A binding site on this subunit recognizes a specific nucleotide sequence on the m R N A just upstream of the start codon. An initiator t R N A (anticodon = U A C) base pairs with the m R N A start codon (A U G). This t R N A carries the amino acid methionine (Met). G T P goes to G D P plus inorganic phosphate.
2. The arrival of the large subunit completes the translation initiation complex. Proteins called initiation factors (not shown) are required to bring all the translation components together. Hydrolysis of G T P provides the energy for the assembly. The initiator t R N A is in the P site (shown here attached to the amino acid Met); the A site is available to the t R N A bearing the next amino acid.

Animation

Initiation of Translation

image

image

The union of mRNA, initiator tRNA, and the small ribosomal subunit is followed by the attachment of a large ribosomal subunit, completing the *translation initiation complex* (see Figure 17.19). Proteins called *initiation factors* are required to bring all these components together. The cell also expends energy obtained by hydrolysis of a GTP molecule to form the initiation complex. At the completion of the initiation process, the initiator tRNA sits in the P site of the ribosome, and the vacant A site is ready for the next aminoacyl tRNA. Note that a polypeptide is always synthesized in one direction, from the initial methionine at the amino end, also called the N-terminus, toward the final amino acid at the carboxyl end, also called the C-terminus (see Figure 5.15).

Elongation of the Polypeptide Chain

In the elongation stage of translation, amino acids are added one by one to the previous amino acid at the C-terminus of the growing chain. Each addition involves several proteins called *elongation factors* and occurs

in a three-step cycle described in **Figure 17.20**. Energy expenditure occurs in the first and third steps. Codon recognition requires hydrolysis of one molecule of GTP, which increases the accuracy and efficiency of this step. One more GTP is hydrolyzed (broken up) to provide energy for the translocation step.

Figure 17.20

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The elongation cycle of translation.
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The elongation cycle of translation.

The hydrolysis of GTP plays an important role in the elongation process; elongation factors are not shown.

The steps are as follows.

1. Codon recognition. Messenger R N A is in the ribosome between the subunits. The anticodon of an incoming aminoacyl t R N A base-pairs with the complementary m R N A codon in the A site. In the P site is a t R N A adding to the growing polypeptide chain. The E site in this diagram is empty. G T P is hydrolyzed to G D P plus inorganic phosphate, increasing the accuracy and efficiency of this step. Although not shown, many different aminoacyl t R N As are present, but only the one with the appropriate anticodon will bind and allow the cycle to progress.
2. Peptide bond formation. An rR N A molecule of the large ribosomal subunit catalyzes the formation of a peptide bond between the amino group of the new amino acid in the A site and the carboxyl end of the growing polypeptide in the P site. This step removes the polypeptide from the t R N A in the P site and attaches it to the amino acid on the t R N A in the A site. E site remains empty in this diagram.
3. Translocation. G T P goes to G D P plus inorganic phosphate. The ribosome translocates the t R N A in the A site to the P site. At the same time, the empty t R N A in the P site is moved to the E site, where it is released. The m R N A moves along with its bound t R N As (arrows show direction of movement to the left, toward the 5 prime end of the m R N A). The movement brings the next codon to be translated to the A site. The ribosome is now ready for the next aminoacyl t R N A.

Animation

Elongation Cycle of Translation

image

image

The mRNA is moved through the ribosome in one direction only, 5′ end first; this is equivalent to the ribosome moving 5′→3′ on the mRNA. The main point is that the ribosome and the mRNA move relative to each other, unidirectionally, codon by codon. The elongation cycle takes less than a tenth of a second in bacteria and is repeated as each amino acid is added until the polypeptide is completed. The empty tRNAs that are released from the E site return to the cytoplasm, where they will be reloaded with the appropriate amino acid (see Figure 17.17).

CheckPoint

Expressing a Gene and Using the Codon Table

Check your understanding before you move on, with the CheckPoint: Expressing a Gene and Using the Codon Table.

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<p>iframes are not supported by your browser.</p>
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Termination of Translation

The final stage of translation is termination **(Figure 17.21)**. Elongation continues until a stop codon in the mRNA reaches the A site. The nucleotide base triplets UAG, UAA, and UGA (all written 5′→3′) do not code for amino acids but instead act as signals to stop translation. A *release factor*, a protein shaped like an aminoacyl tRNA, binds directly to the stop codon in the A site. The release factor causes the addition of a water molecule instead of an amino acid to the polypeptide chain. (Water molecules are abundant in the cytosol.) This reaction hydrolyzes (breaks) the bond between the completed polypeptide and the tRNA in the P site, releasing the polypeptide through the exit tunnel of the ribosome's large subunit. The remainder of the translation assembly then comes apart in a multistep process, aided by other protein factors. Breakdown of the translation assembly requires the hydrolysis of two more GTP molecules.

Figure 17.21

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The termination of translation.
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The termination of translation.

Like elongation, termination requires GTP hydrolysis as well as additional protein factors, which are not shown here.

The steps are as follows.

1. Ribosome reaches a stop codon on the m R N A. The A site accepts a release factor (protein shaped like a t R N A) instead of an amino acyl t R N A. An m R N A is drawn between ribosomal subunits, with a t R N A in the P site attached to a polypeptide chain. The ribosome reaches a stop codon on the m R N A (U A G, U A A or U G A) and the ribosome A site accepts a release factor (protein shaped like a t R N A) instead of an amino acyl t R N A.
2. The release factor promotes hydrolysis of the bond between the t R N A in the P site and the last amino acid of the polypeptide, freeing the polypeptide from the ribosome. The t R N A that had been in the P site is also released from the ribosome in this step.
3. G T P goes to G D P + inorganic phosphate. The two ribosomal subunits and the other components of the assembly disassociate (come apart) with the hydrolysis of two G T P molecules.

Animation

Termination of Translation

image

image

Completing and Targeting the Functional Protein

The process of translation is often not sufficient to make a functional protein. In this section, you will learn about modifications that polypeptide chains undergo after the translation process as well as some of the mechanisms used to target completed proteins to specific sites in the cell.

Protein Folding and Post-translational Modifications

During its synthesis, a polypeptide chain begins to coil and fold spontaneously as a consequence of its amino acid sequence (primary structure), forming a protein with a specific shape: a three-dimensional molecule with secondary and tertiary structure (see Figure 5.18). Thus, a gene determines primary structure, which in turn determines shape.

Additional steps—*post-translational modifications*—may be required before the protein can begin doing its particular job in the cell. Certain amino acids may be chemically modified by the attachment of sugars, lipids, phosphate groups, or other additions. Enzymes may remove one or more amino acids from the leading (amino) end of the polypeptide chain. In some cases, a polypeptide chain may be enzymatically cleaved into two or more pieces. In other cases, two or more polypeptides that are synthesized separately may come together, if the protein has quaternary structure; an example is hemoglobin (see Figure 5.18).

BioFlix® Animation

Protein Processing

image

image

Targeting Polypeptides to Specific Locations

In electron micrographs of eukaryotic cells active in protein synthesis, two populations of ribosomes are evident: free and bound (see Figure 6.10). Free ribosomes are suspended in the cytosol and mostly synthesize proteins that stay in the cytosol and function there. In contrast, bound ribosomes are attached to the cytosolic side of the endoplasmic reticulum (ER) or to the nuclear envelope. Bound ribosomes make proteins of the endomembrane system (see Figure 6.15) as well as proteins secreted from the cell, such as insulin. It is important to note that the ribosomes themselves are identical and can alternate between being free ribosomes one time they are used and being bound ribosomes the next.

What determines whether a ribosome is free in the cytosol or bound to rough ER? Polypeptide synthesis always begins in the cytosol as a free ribosome starts to translate an mRNA molecule. There, the process continues to completion—*unless* the growing polypeptide itself cues the ribosome to attach to the ER. The polypeptides of proteins destined for the endomembrane system or for secretion are marked by a **signal peptide**, which targets the protein to the ER **(Figure 17.22)**. The signal peptide, a sequence of about 20 amino acids at or near the leading end (N-terminus) of the polypeptide, is recognized as it emerges from the ribosome by a protein-RNA complex called a **signal-recognition particle (SRP)**. This particle escorts the ribosome to a receptor protein built into the ER membrane. The receptor is part of a multiprotein translocation complex. Polypeptide synthesis continues there, and the growing polypeptide snakes across the membrane into the ER lumen via a protein pore. The signal peptide is removed by an enzyme. The rest of the completed polypeptide, if it is to be secreted from the cell, is released into solution within the ER lumen (see Figure 17.22). Or, if the polypeptide is to be a membrane protein, amino acid sequences further in the chain cause that part to remain embedded in the ER membrane. In either case, it travels in a transport vesicle to its destination (see Figure 7.9).

Figure 17.22

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The signal mechanism for targeting proteins to the ER.
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The signal mechanism for targeting proteins to the E R.

The steps are as follows.

1. Polypeptide synthesis begins on a free ribosome in the cytosol.
2. An S R P binds to the signal peptide, halting synthesis momentarily.
3. The S R P binds to a receptor protein in the ER membrane, part of a protein complex that forms a pore and has a signal-cleaving enzyme.
4. The S R P leaves, and polypeptide synthesis resumes, with simultaneous translocation across the membrane.
5. The signal-cleaving enzyme cuts off the signal peptide.
6. The rest of the completed polypeptide leaves the ribosome and folds into its final conformation. Ribosomal components disassociate.

MAKE CONNECTIONS *If this protein were destined for secretion, what would happen to it after its synthesis was completed? See*Figure 7.9.**

*For suggested answer, see*Appendix A.**

Animation

Cotranslational Translocation

image

Other kinds of signal peptides are used to target polypeptides to mitochondria, chloroplasts, the interior of the nucleus, and other organelles that are not part of the endomembrane system. The critical difference in these cases is that translation is completed in the cytosol before the polypeptide is imported into the organelle. Translocation mechanisms also vary, but in all cases studied to date, the “postal zip codes” that address proteins for secretion or to cellular locations are signal peptides of some sort. Bacteria also employ signal peptides to target proteins to the plasma membrane or for secretion.

Making Multiple Polypeptides in Bacteria and Eukaryotes

In previous sections, you learned how a single polypeptide is synthesized using the information encoded in an mRNA molecule. When a polypeptide is required in a cell, though, the need is for many copies, not just one.

In both bacteria and eukaryotes, multiple ribosomes translate an mRNA at the same time **(Figure 17.23)**; that is, a single mRNA is used to make many copies of a polypeptide simultaneously. Once a ribosome is far

enough past the start codon, a second ribosome can attach to the mRNA, eventually resulting in a number of ribosomes trailing along the mRNA. Such strings of ribosomes, called **polyribosomes** (or **polysomes**), can be seen with an electron microscope; they can be either free or bound. They enable a cell to rapidly make many copies of a polypeptide.

Figure 17.23

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Polyribosomes.
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Polyribosomes are diagrammed.

Part a. An m R N A molecule is generally translated simultaneously by several ribosomes in clusters called polyribosomes. An m R N A with 5 prime end (start) at the left of a polyribosome (here, cluster of 4 ribosomes) and 3 prime end at the left (end of m R N A). The ribosome at the 3 prime end has completed translating a polypeptide; ribosomes that have completed translation disassociate. Ribosomes not yet at the 3 prime end of the m R N A are still synthesizing polypeptides. Incoming ribosomal subunits will begin translation at the 5 prime end of the m R N A.

Part b. Micrograph of a large polyribosome in a bacterial cell. Ribosomes appear as beads along a string (m R N A). Scale: 0.1 micrometers.

Another way both bacteria and eukaryotes increase the number of copies of a polypeptide is by transcribing multiple mRNAs from the same gene. However, the coordination of the two processes—transcription and translation—differs in the two groups. The most important differences between bacteria and eukaryotes arise from the bacterial cell's lack of compartmental organization. Like a one-room workshop, a bacterial cell ensures a streamlined operation by coupling the two processes. With no nuclear envelope, it can simultaneously transcribe and translate the same gene **(Figure 17.24)**, and the newly made protein can quickly diffuse to its site of function.

Figure 17.24

Coupled transcription and translation in bacteria.

Coupled transcription and translation in bacteria. Micrograph (TEM) of strand of coupled transcription and translation in E. coli D N A (scale: 0.25 micrometers), with diagram detailing this process.

Coupled transcription and translation in bacteria. Micrograph (scale: 0.25 micrometers) shows D N A, m R N A, R N A polymerases, and polyribosomes. In the diagram, four R N A polymerases transcribe a strand of D N A. Attached to each R N A polymerase is a polyribosome (cluster of ribosomes), with each ribosome in

the cluster simultaneously translating an m R N A into a polypeptide as the m R N A is transcribed. (Here, a polypeptide protrudes from each ribosome in the polyribosome). Arrows indicate transcription of the D N A occurs from left to right.

In bacterial cells, the translation of mRNA can begin as soon as the leading 5′ end of the mRNA molecule peels away from the DNA template. The micrograph (TEM) shows a strand of *E. coli* DNA being transcribed by RNA polymerase molecules. Attached to each RNA polymerase molecule is a growing strand of mRNA, which is already being translated by ribosomes. The newly synthesized polypeptides are not visible in the micrograph but are shown in the diagram.

VISUAL SKILLS *Which one of the mRNA molecules started being transcribed first? On that mRNA, which ribosome started translating the mRNA first?*

*For suggested answer, see*Appendix A*.*

In contrast, the eukaryotic cell's nuclear envelope segregates transcription from translation and provides a compartment for extensive RNA processing. This processing stage includes additional steps, discussed earlier, the regulation of which can help coordinate the eukaryotic cell's elaborate activities. **Figure 17.25** summarizes the path from gene to polypeptide in a eukaryotic cell.

Figure 17.25

A summary of transcription and translation in a eukaryotic cell.

A summary of transcription and translation in a eukaryotic cell.

The steps are as follows.

1. Transcription. Inside the nucleus, R N A is transcribed from D N A template. R N A transcript shown with 3 prime end inside R N A polymerase on the D N A template and 5 prime end completed.
2. R N A processing. R N A transcript (pre-m R N A) with introns (longer segments) and exons (shorter segments) labeled. In the eukaryote nucleus, the R N A transcript (pre-m R N A) is spliced and modified to produce m R N A, which moves from the nucleus to the cytoplasm.
3. An m R N A (introns spliced out) leaves the nucleus. The 5 prime cap is in cytoplasm; Poly-A tail still inside nucleus. Arrows indicate m R N A will move completely out of the nucleus and attach to the small ribosomal subunit, with the 5 prime cap at the left. Large ribosomal subunit will attach, with the m R N A between the two units. Sites E, P and A are labeled on the large ribosomal subunit.

4. Amino Acid Activation. Each amino acid attaches to its proper t R N A with the help of a specific enzyme and A T P. Here, a t R N A is shown inside aminoacyl-t R N A synthase and its amino acid is being attached. Other t R N As and amino acids are also drawn free in the cytoplasm, not yet attached.
5. Translation. A succession of t R N As add their amino acids to the polypeptide chain as the m R N A is moved through the ribosome one codon at a time. When completed, the polypeptide is released from the ribosome.

Here, the m R N A strand in the ribosome has the sequence 5 prime-U G G U U U A U G-3 prime. In the P (middle) site, a t R N A with the anticodon A A A binds to the m R N A codon U U U, adding its amino acid to the growing polypeptide chain.

Another t R N A with anticodon U A C and a different amino acid is entering the A (right) site, where the m R N A has codon A U G.

A third t R N A is exiting from the E (left) site. This t R N A has anticodon A C C and the m R N A codon it has already translate (transferred its amino acid to the polypeptide chain) was U G G.

This diagram shows the path from one gene to one polypeptide. Each gene in the DNA can be transcribed repeatedly into many identical RNA molecules, and each mRNA can be translated repeatedly to yield many identical polypeptide molecules. (Remember that the final products of some genes are not polypeptides but RNA molecules that don’t get translated, including tRNAs and rRNAs.) In general, the steps of transcription and translation are similar in bacterial, archaeal, and eukaryotic cells. The major difference is the occurrence of RNA processing in the eukaryotic nucleus. Other significant differences are found in the initiation stages of both transcription and translation and in the termination of transcription. To visualize these processes in their cellular context, see Figure 6.32d-f.

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